

PlayIQ Module 5: Compression Learning

Your mission: make a big idea small without breaking it

Imagine being able to look at a crowded chapter, a long video, or a page of notes and pull out the few ideas that actually hold everything together.

That is compression. It is not shrinking words just to make a shorter page. It is building a clear mental map that keeps the truth, removes clutter, and helps you remember what matters.

By the end of this module, I can:

- separate a topic's core idea from its supporting details;
- simplify an explanation without making it inaccurate;
- explain one idea in three useful ways;
- build a one-page Understanding Card;
- compare audio review, a visual card, and teach-back fairly;
- choose a memory-review setting for Orion to test;
- create evidence and a knowledge-file record for my personal tutor Project.

ORION: "A powerful question is not always longer. Sometimes it removes

everything that does not matter yet. Today, you will learn how to fold a big

topic into a map small enough to carry—and accurate enough to trust."

Part 0: The 45-second compression hook

Your friend missed a lesson called **Why We Have Seasons** and texts:

"I have 30 seconds. What is the one idea I need first?"

Choose the strongest first reply:

- ☐ "Earth has weather, temperatures, months, hemispheres, an orbit, an axis, sunlight, and different seasons."
- ☐ "Seasons mainly happen because Earth's tilted axis changes the angle and length of sunlight each hemisphere receives during the year."
- ☐ "Summer is hot and winter is cold."

My quick choice: _____

Why it works better than the other two:

The first answer has too many loose details. The third is short but does not explain the cause. The second is compressed: it is smaller than the full lesson but keeps the central relationship.

That is today's challenge—**short enough to use, accurate enough to trust.**

Part 1: Compression is not deletion

A useful compression answers three questions:

1. What is the central idea?
2. Which details are necessary to understand it?
3. What would become false or misleading if removed?

Weak compression

"Photosynthesis is plants eating sunlight."

It is memorable, but it is inaccurate. Plants do not eat sunlight. The phrase also removes water, carbon dioxide, and the creation of sugar.

Safer compression

"Photosynthesis is how plants use light energy, water, and carbon dioxide to

make sugar, releasing oxygen as part of the process."

This version is still simple, but it preserves the major ingredients, action, and result.

Orion's truth check

STUDENT: "Can you make this as short as possible?"

ORION: "I can make it shorter, but first tell me what you need it for. A one-line preview, a test explanation, and a lab report require different levels of detail."

STUDENT: "I need a first map before studying the chapter."

ORION: "Then I will give you the core idea, three necessary parts, and one warning about what the short version leaves out."

Why this matters: the purpose controls how much detail is safe to remove.

My compression purpose

Choose one real topic from school or a parent-approved learning source.

Topic: _____

Source: _____

I need the compression for:

- ☐ a first preview
- ☐ a study review
- ☐ explaining it to someone else
- ☐ preparing for a question or assignment
- ☐ another purpose: _____

What must remain accurate for that purpose?

Part 2: Build the compression ladder

The Compression Ladder turns one large topic into four increasingly useful levels.

Level 1: The topic in one question

What is the main question this topic answers?

Example: "How do plants turn light energy into stored chemical energy?"

My topic's main question:

Level 2: The one-sentence core

Answer the question in one accurate sentence. Do not try to include everything.

My one-sentence core:

Level 3: The three supporting pillars

Choose the three details someone needs before the core idea makes sense.

1. _____

2. _____

3. _____

Level 4: The accuracy guardrail

Write one warning about what your short explanation leaves out or could make someone misunderstand.

Example: "This summary leaves out the detailed chemical stages and should not be used as the complete explanation of photosynthesis."

My accuracy guardrail:

Why this activity matters

The ladder lets you make a topic smaller while leaving a path back to the full source. Your future tutor should never confuse a quick map with complete knowledge.

Part 3: The 3-Ways Rule

Understanding becomes stronger when you can move an idea between forms. Use the same core idea in three ways.

Way 1: Big-picture explanation

State what the whole idea does and why it matters.

Example:

"Supply and demand describes how the amount people want and the amount

available can influence price and quantity in a market."

My big-picture explanation:

Way 2: Step-by-step explanation

Show the sequence or cause-and-effect chain.

Example:

"A popular item becomes scarce. More buyers compete for fewer items. Sellers

can raise the price. The higher price may reduce demand or encourage more

supply."

My step-by-step explanation:

1. _____
2. _____
3. _____
4. _____

Way 3: Example-first explanation

Use a specific case, then connect it to the rule.

Example:

"If only 50 concert tickets remain and 500 people want them, the tickets may

become more expensive. That is one example of high demand meeting limited

supply."

My example-first explanation:

Accuracy check

For each version, ask:

- Does it preserve the central relationship?
- Are important terms used correctly?
- Does the example actually match the rule?
- Did I make a statement broader than the source supports?
- What detail did I intentionally leave out?

One correction I made after checking:

ORION: "If you can only repeat one wording, the idea may still belong to

the page. If you can explain it accurately in several forms, it is beginning

to belong to you."

Part 4: Learn How You Learn discovery

Answer honestly. These questions help Orion design tests for your review process; they do not assign you a learning-style identity.

1. When a chapter looks too large, what makes you feel ready to begin?

2. Which is harder for you: deciding what matters or deciding what can be left out? Why?

3. How can you tell the difference between recognizing an explanation and being able to produce it yourself?

4. When have you remembered something because you explained it to another person?

5. When has a diagram, card, or page layout helped you see a relationship?

6. If an audio explanation sounds clear in the moment, how could you test what remained after listening?

Part 5: Build a one-page Understanding Card

Use your Compression Ladder and 3-Ways explanations to complete this card.

```
=====
ONE-PAGE UNDERSTANDING CARD
=====
TOPIC: _____
```

PURPOSE: _____

SOURCE: _____

MAIN QUESTION

ONE-SENTENCE CORE

THREE SUPPORTING PILLARS

1. _____

2. _____

3. _____

ONE EXAMPLE

KEY TERMS

1. _____ = _____

2. _____ = _____

3. _____ = _____

COMMON MISTAKE OR MISCONCEPTION

ACCURACY GUARDRAIL—WHAT THIS SHORT VERSION LEAVES OUT

TWO RETRIEVAL QUESTIONS

1. _____

2. _____

=====

Orion review prompt

"Review my Understanding Card against the source I provide. Identify one

sentence that is accurate, one place that may be vague or misleading, and

one important detail I may have removed. Do not rewrite the whole card. Ask

me to make the correction."

Correction I chose to make:

Why it improves accuracy:

Part 6: Active comparison—three review formats

Now compare three ways of reviewing the **same Understanding Card**. The goal is to see what helps you retrieve the idea, not what merely feels comfortable.

Prepare a fair test

Use these five checks after each method:

1. State the one-sentence core.
2. Name the three pillars.
3. Give one example.
4. Explain the common mistake.
5. State the accuracy guardrail.

Score one point for each answer you can produce without looking. If an answer is partly correct, mark it **partial** and describe what was missing instead of inventing a decimal score.

Method A: Audio review

With parent approval, read the card aloud or use an approved text-to-speech feature. Listen once. Put the card away, take a short neutral break, then answer the five checks.

What I remembered without looking:

Correct answers: ____ / 5 Partial answers: ____ Not yet: ____

What audio made easier or harder:

Method B: Visual card review

Study the card as a page for the same amount of time. Cover it, take the same kind of break, then answer the five checks.

What I remembered without looking:

Correct answers: ____ / 5 Partial answers: ____ Not yet: ____

What the card layout made easier or harder:

Method C: Teach-back review

Explain the card aloud to a parent, peer, camera, or voice recorder approved by your parent. If no person or recording is available, teach an empty chair. Do not look at the card while explaining. Then answer the five checks.

What I remembered without looking:

Correct answers: ____ / 5 Partial answers: ____ Not yet: ____

What teach-back made easier or harder:

Fairness checklist

- ☐ I used the same topic and five checks.
- ☐ Each review received about the same amount of time.

- [] I answered before checking the source.
- [] I recorded partial or incorrect answers honestly.
- [] I did not ask Orion to pretend that I remembered something.

Part 7: Evidence interpretation

Method	Correct / 5	Partial	Frustration after (1–5)	Could explain the core?	Useful for this task?
Audio	—	—	—	Yes / Partly / Not yet	Yes / Maybe / No
Visual card	—	—	—	Yes / Partly / Not yet	Yes / Maybe / No
Teach-back	—	—	—	Yes / Partly / Not yet	Yes / Maybe / No

Which method supported the strongest retrieval in this experiment?

What exact evidence supports that conclusion?

Which method felt easy but did not produce strong recall, if any?

Which method may be useful for a different purpose?

What could have influenced the result besides the format?

Use conditional language

Avoid: "I am a visual learner."

Prefer:

"When reviewing this kind of topic, the visual card helped me retrieve

_____, while teach-back exposed _____. I will test _____ first

on another topic before making it a stronger tutor rule."

My interpretation:

Part 8: Learning Supercharger Blueprint update

The Blueprint is a collection of tutor settings you and Orion test together.

This module's setting tells Orion how to begin a review session. It does not declare one format best for every subject or every day.

Proposed tutor setting

review-start = [audio | visual-card | teach-back | mixed | not-yet-known]

use-when = _____

evidence = _____

accuracy-check = "Ask me to retrieve before showing the source."

fallback = _____

Write it in plain language:

"When I review _____, begin with _____. Then ask

me to explain the core idea without looking. If my retrieval is incomplete,

use _____ and test me again."

Rule status:

- [] `experiment\_result`
- [] `student\_reported`
- [] `pending\_validation`

Part 9: Structured Knowledge File update

Add the following to `Learning_Rules.md` or the approved equivalent. Keep the source and uncertainty visible so Orion does not turn one activity into a fixed trait.

```
- rule_id: module_5_review_start
module: 5
skill_tested: compression_and_retrieval
date: "[YYYY-MM-DD]"
source_activity: "audio_vs_visual_card_vs_teach_back"
topic: "[topic]"
compression_purpose: "[preview|study_review|teach_someone|assignment_prep|other]"
proposed_review_start: "[method or not_yet_known]"
evidence:
audio_result: "[correct/partial/not_yet observations]"
visual_card_result: "[correct/partial/not_yet observations]"
teach_back_result: "[correct/partial/not_yet observations]"
accuracy_correction: "[correction made to card]"
provenance: "[experiment_result|student_reported|pending_validation]"
confidence: "[low|medium|high]"
fallback_method: "[method or pending_validation]"
privacy: "safe_for_tutor_project"
review_condition: "Retest on another subject or when retrieval remains weak."
```

Ask Orion:

"Read my Module 5 record. Tell me what I demonstrated, what is still only a

preference, and what should be tested again. Preserve my wording and do not

invent a memory score."

What Orion identified as still uncertain:

Part 10: Proof Artifact—Retention Test Card

=====

PLAYIQ RETENTION TEST CARD

=====

Student: _____ Date: _____

Topic compressed: _____

Compression purpose: _____

Source checked: _____

My one-sentence core:

Accuracy correction I made:

REVIEW COMPARISON

Audio: correct ____ / 5 | partial ____ | frustration ____ / 5

Visual card: correct ____ / 5 | partial ____ | frustration ____ / 5

Teach-back: correct ____ / 5 | partial ____ | frustration ____ / 5

Proposed `review-start`: _____

Evidence: _____

Fallback: _____

Status: [experiment_result | student_reported | pending_validation]

=====

Keep the Understanding Card and Retention Test Card. They will help Orion learn
how your personal tutor should compress, review, and verify information.

Part 11: Mastery Challenge—teach it three ways

Choose a real concept that matters in your current learning. Use a school-approved source.

1. Build a Compression Ladder.
2. Complete the one-page Understanding Card.
3. Ask Orion to flag possible inaccuracies without rewriting it.
4. Correct the card yourself.
5. Teach the concept in all three forms below.

Form A: Big picture

Explain what the concept is and why it matters in two or three sentences.

Form B: Steps or cause-and-effect

1.

2.

3.

4.

Form C: Example first

Accuracy defense

What important detail did you choose to keep, and why?

What did you leave out because it was not necessary for this purpose?

How did you confirm that your short explanation was still accurate?

Mastery check

- ☐ My compression answers the topic's main question.
- ☐ My three versions preserve the same core idea.
- ☐ I used an approved source to check accuracy.
- ☐ I identified what the short version leaves out.
- ☐ I retrieved the idea instead of only rereading it.
- ☐ I saved the proof artifact and provenance record.

ORION: "You can now make a large topic usable without pretending the

short version is the whole truth. In Module 6, you will test what remained,

turn mistakes into clues, and build a review plan from real evidence."

Beta tester feedback—complete only in the beta route

This section appears only after the module is complete in the beta-testing route. Standard-course students do not complete it.

1. Which part made compression feel useful or satisfying?

2. Which instruction, template, or example was confusing?

3. Did the 30–60 second hook feel fast and fun? Why or why not?

4. Did the three-method comparison feel fair and possible to complete?

5. What would add more excitement or usefulness without adding unnecessary work?

Confidential - PlayIQ Academy & Orion AI © 2026